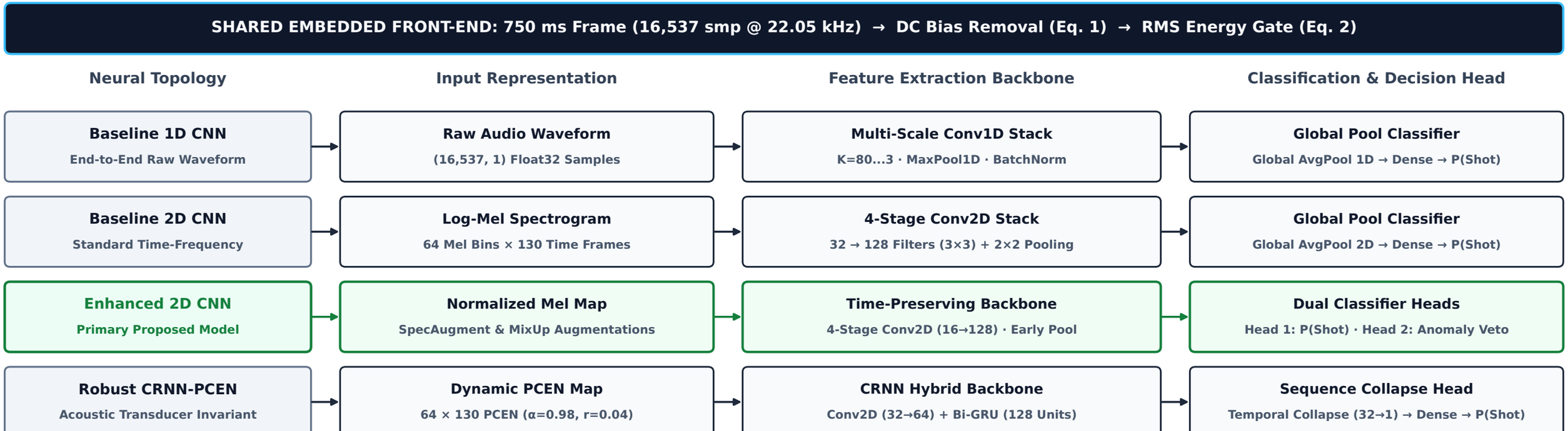
SHARED EMBEDDED FRONT-END: 750 ms Frame (16,537 smp @ 22.05 kHz) → DC Bias Removal (Eq. 1) → RMS Energy Gate (Eq. 2)
Neural Topology
Input Representation
Feature Extraction Backbone
Classification & Decision Head
Baseline 1D CNN
End-to-End Raw Waveform
Raw Audio Waveform
(16,537, 1) Float32 Samples
Multi-Scale Conv1D Stack
K=80...3 · MaxPool1D · BatchNorm
Global Pool Classifier
Global AvgPool 1D → Dense → P(Shot)
Baseline 2D CNN
Standard Time-Frequency
Log-Mel Spectrogram
64 Mel Bins × 130 Time Frames
4-Stage Conv2D Stack
32 → 128 Filters (3×3) + 2×2 Pooling
Global Pool Classifier
Global AvgPool 2D → Dense → P(Shot)
Enhanced 2D CNN
Primary Proposed Model
Normalized Mel Map
SpecAugment & MixUp Augmentations
Time-Preserving Backbone
4-Stage Conv2D (16→128) · Early Pool
Dual Classifier Heads
Head 1: P(Shot) · Head 2: Anomaly Veto
Robust CRNN-PCEN
Acoustic Transducer Invariant
Dynamic PCEN Map
64 × 130 PCEN (α=0.98, r=0.04)
CRNN Hybrid Backbone
Conv2D (32→64) + Bi-GRU (128 Units)
Sequence Collapse Head
Temporal Collapse (32→1) → Dense → P(Shot)